

IdeaStamp — course prototype

Minimal end-to-end flow from the Assignment 2 white paper: on-chain salted commitments, optional reveal, and public verification. Token staking, DAO, and extended dispute flows are out of scope for this UI prototype.

Commit

Pay commit fee and submit commitmentHash

Reveal

Reveal contentHash and salt

Verify

Derive or look up a commitment on-chain

My Records

Commitments for the connected wallet

Local chain: run `npx hardhat node`, then `npm run deploy:local` (updates `frontend/.env.local`).

Commit

Hash text or file bytes with Keccak-256 locally, combine with salt and metadata hash per the contract formula, then pay `commitFeeWei` to submit on-chain.

☐ Use a file as content (otherwise use the text area below)

Idea / draft text

test

Metadata (optional)

Leave empty for metadataHash = bytes32(0)

Salt

☐ Random 32 bytes ☒ Text (hashed to bytes32) ☐ Raw bytes32 hex

kevin

CID (optional)

ipfs://... or leave empty

contentHash: 0x9c22ff5f21f0b81b113e63f7db6da94fedef11b2119b4088b89664fb9a3cb658
metadataHash: 0x00
salt: 0x36b3d9a3a5f553e3928d64b60e46277b23b2d28a3c41570aa33984208cab1df1
commitmentHash: 0x07b5ce16118710683189595f893a909a5a52dd4d4c76c7ad262862083ac1ff18
commitFeeWei: 100000000000000

Submit commit

Reveal

Use the same content/file, metadata, and salt as at commit time; chain ID and author address must match.

☐ Use file

test

Metadata (empty = bytes32(0))

☒ Salt as text (keccak256 utf8) ☐ Salt as bytes32 hex

kevin

Reveal CID (optional)

commitment: 0x07b5ce16118710683189595f893a909a5a52dd4d4c76c7ad262862083ac1ff18

reveal

tx: 0x0289e47c39f1a5171e6716e17ff0fc08bef6d3c970898fd6b93a2f1a08ce1fd8

Revealed. Keep salt and content for verifiers: salt =
0x36b3d9a3a5f553e3928d64b60e46277b23b2d28a3c41570aa33984208cab1df1

Verify

Derive the commitment from author + content + salt locally, or paste a commitment hash, then read on-chain state (wallet RPC must point at the deployed network).

☒ Derive from content ☐ Paste commitment hash

Author address (empty = connected wallet: 0xf39Fd6e51aad88F6F4ce6aB8827279cfffB92

☐ Use file

test

Metadata (optional)

☒ Salt: text ☐ Salt: hex

kevin

Current chain ID (from wallet): 31337 – derivation uses this chainId.

Lookup hash: 0x07b5ce16118710683189595f893a909a5a52dd4d4c76c7ad262862083ac1ff18

On-chain record. Revealed: yes

author: 0xf39Fd6e51aad88F6F4ce6aB8827279cfffB92266

committedAt: 1775339518

committedBlock: 3

commitCid: –

revealedContentHash: 0x9c22ff5f21f0b81b113e63f7db6da94fedef11b2119b4088b89664fb9a3cb658

revealSalt: 0x36b3d9a3a5f553e3928d64b60e46277b23b2d28a3c41570aa33984208cab1df1

revealCid: –

My Records

0xf39Fd6e51aad88F6F4ce6aB8827279cfffB92266

0x07b5ce16118710683189595f893a909a5a52dd4d4c76c7ad262862083ac1ff18

committedAt: 1775339518

block: 3

revealed: yes

commitCid: —